Avishkar AV

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PROFESSIONAL SUMMARY

Fullstack Software Engineer with production experience architecting high-concurrency microservices, geospatial data engines, and scalable cloud platforms. Sole backend architect of bluedot.ngo (FastAPI, PostGIS, Cashfree 80G tax compliance) and core engineer of Passiton serving 16,000+ active users on GCP Cloud Run. Proven track record in database performance tuning, Redis in-memory caching, automated CI/CD pipelines, and regulatory data compliance (DPDP/GDPR).

TECHNICAL SKILLS

Languages & Backend: Python, SQL, JavaScript, TypeScript, Dart, Java, FastAPI, Flask, SQLAlchemy 2.0, Alembic, Pydantic, REST APIs, JWT, RBAC

Databases & Cloud: PostgreSQL, PostGIS (ST_Contains, ST_DWithin, SRID 4326), Redis, Supabase, MySQL, MongoDB, GCP (Cloud Run, Vertex AI), AWS (EC2), Docker, CI/CD

FinTech, APIs & Tools: Cashfree Payment Gateway (Webhooks, 80G Tax Receipts), YouTube Data API v3, MSG91, OpenStreetMap, Git, Linux, Postman

WORK EXPERIENCE

Fullstack Developer, Third Eye Creative Dec 2025 – Present | Bengaluru, India

- **Passiton (16,000+ Active Users):** Maintained production recycling marketplace on GCP Cloud Run with autoscaling; architected dual ASGI server running FastAPI for mobile and Flask for admin under a unified container.
- **BlueDot (Solo Backend Architect):** Architected entire backend for bluedot.ngo as an async FastAPI REST API on PostgreSQL (Supabase) with SQLAlchemy 2.0 ORM, Pydantic schemas, and Alembic migrations.
- **Database Performance Tuning:** Shrank production PostgreSQL footprint **60% (2GB to 800MB)** and accelerated complex join queries **40%** through zero-downtime poison-pill row cleanup and targeted indexing.
- **Latency & Cost Optimization:** Slashed feed response latency from **10s to <2s** with a Redis-cached feed algorithm; eliminated 100% of third-party map costs via OpenStreetMap.
- **PostGIS Geospatial Engine:** Stored restoration boundary polygons (SRID 4326) and built automated PostgreSQL triggers with ST_Contains to increment cached tree counts without runtime aggregates.
- **FinTech & 80G Tax Compliance:** Integrated Cashfree Payment Gateway supporting webhook signature verification, delayed PAN collection UX flow, and automated 80G tax receipt PDF generation.
- **AI Reliability Engineering:** Cut Vertex AI API failure rate from **33% to 3%** with exponential backoff retry logic; integrated dual-engine AI vision (Vertex AI & Pl@ntNet) with 70% threshold auto-publishing.
- **Cloud Infrastructure & CI/CD:** Cut infrastructure spend **46%** by migrating legacy cash-collection tool from AWS EC2 to DigitalOcean droplets with Docker; built multi-tenant CI/CD with GitHub Actions.

Fullstack Developer Intern, EarthlyExotica Jan 2025 – May 2025 | Bengaluru, India

- Built mobile-responsive frontend in Flutter for baadige.com utilizing modular UI components; integrated core REST APIs connecting payment flows and product catalog services.

TECHNICAL PROJECTS

- **Autonomous YouTube Video Publishing Engine:** Architected a zero-cost serverless media pipeline auto-publishing daily 1080p Shorts to YouTube via GitHub Actions CI/CD; automated OAuth 2.0 refresh-token handshake and resumable chunked video uploads (PUT protocol) using **16% of daily free quota (1,600 / 10,000 units)**, paired with Gemini Flash AI scriptwriting, local Kokoro-82M ONNX TTS, Faster-Whisper alignment, and 14 Mbps broadcast master headless Chrome CDP rendering.
- **Smart Attendance System:** Real-time facial recognition pipeline on CCTV feeds that replaced manual logging with automated CSV attendance tracking using Python, OpenCV, and NumPy.
- **Weather Alert & SMS Notification System:** Automated daemon broadcasting localized rainfall alerts via SMS from real-time meteorological sensor feeds using Python, OpenWeatherMap API, and Twilio API.

EDUCATION & CERTIFICATIONS

B.E. in Computer Science & Engineering, Visvesvaraya Technological University (VTU) Dec 2021 – June 2025 | CGPA: 7.95 / 10.0

- **Certifications:** Oracle Cloud Infrastructure 2025 Certified Generative AI Professional • Oracle OCI AI Foundations Associate • Java OOP Certification